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Batch: 15

Experiment-48

Algorithm

Algorithm: Find Pivot Index of an Array

1. Start.
2. Read the size of the array `n`.
3. If `n <= 0`, display **"Invalid array size"** and stop.
4. Declare a variable-length integer array `arr[n]`.
5. Read `n` elements into the array.
6. Pass the address of `arr` and `n` to the function `pivot_index()`.
7. In `pivot_index()`:
    - Repeat for each index `i` from `0` to `n-1`:
        - Set `leftSum = 0`.
        - Set `rightSum = 0`.
        - Calculate `leftSum` by adding all elements from index `0` to `i-1`.
        - Calculate `rightSum` by adding all elements from index `i+1` to `n-1`.
        - For the first element, `leftSum` automatically remains `0`.
        - For the last element, `rightSum` automatically remains `0`.
        - If `leftSum == rightSum`:
            - Display `i` as the pivot index.
            - Return from the function so that the **leftmost pivot index** is displayed.
8. If no index satisfies `leftSum == rightSum`, display `-1`.
9. Stop.

Test Case Table

Test Case	Size `n`	Array Elements	Expected Output	Reason
1	6	`1 7 3 6 5 6`	`Pivot index = 3`	Left sum = 11, Right sum = 11
2	3	`1 2 3`	`-1`	No pivot index exists
3	3	`2 1 -1`	`Pivot index = 0`	First index: Left sum = 0, Right sum = 0
4	3	`1 -1 2`	`Pivot index = 2`	Last index: Left sum = 0, Right sum = 0
5	5	`0 0 0 0 0`	`Pivot index = 0`	Every index is a pivot, but the leftmost is index 0

Test Case	Size n	Array Elements	Expected Output	Reason
6	1	10	Pivot index = 0	Left sum = 0 and Right sum = 0
7	0	—	Invalid array size.	Array size must be positive

C Program

```
#include <stdio.h>

void pivot_index(int *arr, int n)
{
   int i, j;
   int leftSum, rightSum;

   for (i = 0; i < n; i++)
   {
      leftSum = 0;
      rightSum = 0;

      // Calculate sum of elements to the left
      for (j = 0; j < i; j++)
      {
         leftSum += *(arr + j);
      }

      // Calculate sum of elements to the right
      for (j = i + 1; j < n; j++)
      {
         rightSum += *(arr + j);
      }

      // Check for pivot index
      if (leftSum == rightSum)
      {
         printf("Pivot index = %d\n", i);
         return;     // Ensures leftmost pivot is displayed
      }
   }

   printf("-1\n");
}

int main()
{
   int n, i;

   printf("Enter the size of array: ");
   scanf("%d", &n);
```

```
    if (n <= 0)
    {
        printf("Invalid array size.\n");
        return 0;
    }

    int arr[n];   // Variable-Length Array

    printf("Enter %d elements: ", n);

    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    // Passing address of array
    pivot_index(arr, n);

    return 0;
}
```

